FINITE POSSIBILITY MECHANICS

MODULAR PAPER 05

Phenomenological Bridges

Route-Cost Correspondences with Quantum, Gravitational, and Cosmological Observables

BRIDGE PAPER

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Contents

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Abstract

Finite Possibility Mechanics supplies one microscopic accounting structure: a bounded directed route ledger, a constitutive viscosity field, a finite action budget, and local signed conservation. This paper develops the physical bridge layer that maps those internal quantities to established observables. The bridges cover pure dephasing, Landauer erasure, gravitational response, time dilation, cosmic acoustic structure, finite Born weights, Bell/CHSH correlations, and a bare electromagnetic coupling. The central result is architectural: the same route-cost variables enter every bridge, so the framework gains unification and accepts shared empirical risk.

The paper distinguishes four operations that are often conflated. An identity follows algebraically after definitions are fixed. A correspondence reproduces the form of an established equation. A calibration fixes dimensional scale from measured input. A prediction evaluates a pre-specified map on data not used to choose it. This classification keeps the five foundational axioms distinct from physical bridge laws that require an additional dimensional, dynamical, or observable dictionary.

> Bridge result
> FPM defines a coherent family of physical hypotheses rather than unrelated curve fits. The five foundational axioms determine the finite accounting structure; temperature, link maintenance, clock progress, and observable maps are separately declared bridge commitments. Tensor galaxy morphology, a complete cosmological likelihood, resource-dependent Bell correlations, and the observable meaning of the finite lag ceiling remain the decisive empirical frontier.

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1. Bridge methodology

1.1 Dependency stack

A physical bridge begins only after the foundational and executable layers are fixed. Paper 1 defines the local ledger and its conservation theorems. Paper 2 defines the real-valued reference runtime. Paper 3 defines the finite carrier, quantization, and information dynamics. Paper 4 defines an exact integer realization. The present paper asks which observable equations can be built from that shared machinery.

$$\text{axioms} \rightarrow \text{route ledger } R \rightarrow \text{viscosity } \Omega \rightarrow \text{action } L \rightarrow \text{carrier } \Psi \rightarrow \text{observable bridge}$$

1.2 Evidence labels

- Derived: follows from stated definitions and earlier results without fitting an additional coefficient.
- Constitutive: a declared physical rule selecting one model from a wider admissible family.
- Calibrated: uses an observed dimensional scale or reference constant.
- Audited: evaluated numerically against an established equation or dataset.
- Predictive: fixed before application to data capable of rejecting it.

A bridge can occupy more than one category. For example, the dephasing recurrence is derived from the selected carrier update and then audited against a Lindblad channel. The gravitational constant calculation is algebraically determined after a temperature and scale map are selected, but its dimensional normalization is calibrated.

1.3 Dimensional discipline

Internal FPM variables are normalized. A bridge to SI observables therefore requires an explicit dimensional dictionary. Every such dictionary must identify its input units, calibration data, domain, and transformation law. Numerical proximity without that dictionary is not a bridge; a bridge is a reproducible map whose assumptions travel with its output.

2. Lindblad correspondence

For the two-state projection developed in Paper 3, the selected diagnostic leaves diagonal probabilities unchanged and multiplies the off-diagonal coherence by a persistence factor kappa. With the ordinary Hamiltonian-driven part held aside, the audited update is

$$\rho_{01,t+1}=\kappa\rho_{01,t},\quad \rho_{00,t+1}=\rho_{00,t},\quad \rho_{11,t+1}=\rho_{11,t},\quad 0\le\kappa\le 1.$$

For kappa=exp(-Gamma Delta t), this is the finite-step pure-dephasing channel generated by the zero-Hamiltonian Lindblad form [5]. The executable audit uses the algebraically equivalent Euler parameterization gamma=(1-kappa)/Delta t. Positivity and trace preservation follow from the dephasing channel itself; no nonzero coherence target is introduced in this audit.

> Status
> This is an algebraic implementation correspondence for the selected zero-target pure-dephasing channel, not independent physical evidence. The physical hypothesis is that the persistence factor is controlled by FPM route burden rather than inserted as an independent environmental decay constant.

The runtime audit compares two implementations of that same recurrence and finds agreement down to machine precision. The discriminating experiment is not another reproduction of the selected curve; it is a test of whether measured decoherence changes with the internal route-cost variables predicted by FPM.

3. Landauer bridge and mass-equivalent scale

Consolidation removes representable alternatives from the finite carrier. The semantic entropy decrease is converted into a thermodynamic debit through Landauer's minimum erasure cost [6]

$$E_{erase} \ge k_B T \ln(2) \text{ per erased bit.}$$

The executable ledger records that debit subject to the energy actually available. Dividing an erasure energy by c squared gives a mass-equivalent scale. Because the finite carrier contains N bit-equivalent slots, repeated erasure defines a discrete ladder rather than a continuous family:

$$m_n=n\, k_B T \ln(2)/c^2,\quad n=0,1,\ldots,N.$$

This ladder is an energy-accounting result. A particle identification additionally requires stable dynamics, symmetry, charge, spin, lifetime, and interaction structure. The carrier capacity N=1,452,997,909 and its shell-filling radius are fixed by the finite construction in Paper 3, so the ladder spacing and endpoint are reproducible once the bridge temperature is specified.

> Status
> The information-to-energy inequality is established thermodynamics; applying it to FPM consolidation is a constitutive identification. The resulting mass-equivalent ladder is exact under that identification. A particle spectrum is a further dynamical problem.

3.1 Candidate nine-channel equipartition state law

The axioms use normalized route cost and do not select a Kelvin-valued temperature. A candidate dimensional extension introduces an energy-per-route dictionary ϵ_R and a thermodynamic state law. The directed 3 by 3 route ledger supplies $n_{eff}=9$ native channels. The additional equipartition postulate assigns the bulk node capacity E_{max} equally across those channels, so $\epsilon_R=h/[n_{eff}\,\Delta t_{univ}]$ and

$$T_{erase}=hE_{max}/[n_{eff}N\,k_B\,\ln(2)\,\Delta t_{univ}]=306.442\text{ K}.$$

A separate canonical edge diagnostic tests whether the first master-chain action state admits a scalar temperature under a route-energy dictionary. Across 375 undirected edges after the tick-zero truth-target update, it gives $\mathrm{corr}(\Delta L, \ln[P_{ij}/P_{ji}])=0.00469041$, $\beta^*= -0.00183473$, and normalized residual 0.99999401. This state therefore fails that proposed canonicality criterion.

The equipartition rule is a proposed constitutive closure, not a consequence of Axioms A1-A5 alone. Substitution into the present dimensional gravitational formula gives $G_{FPM}=6.53990\times10^{-11}\text{ m}^3\text{ kg}^{-1}\text{ s}^{-2}$, 2.0137 percent below the CODATA value. The residual is an unresolved discrepancy, not an inferred screening correction. Any adoption must state the route-energy dictionary, the thermodynamic state law, and the conditions under which no scalar temperature is defined.

4. Emergent gravity and galaxy response

4.1 Route-cost gradients

FPM proposes that matter raises local route burden, route burden changes viscosity, and spatial viscosity gradients deflect propagation. In the continuum bridge, the effective acceleration is written as a baryonic source multiplied by a dimensionless susceptibility nu_FPM. The reference audit uses

$$\nu_{FPM}(x)=1+[\Omega_{max}/\mathrm{sqrt}(x+r_{tensor})]\,[1+E_Z x^2]^{-\beta},$$

$$g_{pred}=\nu_{gb}(x)g_{bar},\quad \nu_{gb}=\max(10^{-9},\nu_{FPM}-\alpha f_{gas}/(1+x)).$$

Here $x=|g_{bar}|/a_0$ is the normalized baryonic-acceleration amplitude, r_{tensor} is the declared tensor-channel contribution, $E_Z=0.20E_{max}$ is the consolidation threshold, and beta is inherited from the viscosity law. The gas fraction is computed at each SPARC radius from the squared velocity contributions, $f_{gas}=\max(0,V_{gas}|V_{gas}|)/V_{bar}^2$, clipped to [0,1]. Thus f_{gas} is observed source composition, not a fitted exponent; alpha=1/5 is the trace-weight coefficient from Paper 1. The low-acceleration inverse-square-root response arises from the matter-load dilution law combined with the network's minimum-connectivity shift.

4.2 SPARC audit

An archived 99-galaxy benchmark record in the source repository [15], constructed from SPARC data [8], stores median velocity RMSE 11.61 km/s for the gas-boundary FPM source functional and 11.7156 km/s for the fixed comparison relation used by the script. The benchmark concerns the modified-dynamics class introduced by Milgrom [7], but the script's exact comparison formula is defined by its archived implementation rather than by the 1983 paper. The repository does not redistribute the SPARC tables or the archived selection and parameter ledger. Therefore these numbers are unregenerated benchmark metadata, not a current empirical result, and they receive no Level 5 evidential weight in the public artifact. When the local dataset path is absent, the runtime emits an unavailable status rather than a competitive verdict. A third-party reproduction must supply the cited data, exact selection list, parameter ledger, source revision, and regenerated outputs before making an empirical comparison claim.

> Empirical obligation
> The scalar rotation-curve audit must be followed by the harder tensor test: the full three-dimensional route ledger must predict non-axisymmetric morphology, lensing, and environmental dependence with one shared parameter ledger.

4.3 Dimensional gravitational scale

The implemented dimensional map is the following explicit chain. It uses the action ceiling through L_{max}, exact carrier capacity N, carrier radius α_{PP} derived by the shell fixed point in Paper 3 Section 6.1, Planck constant h, electron mass m_e, c, k_B, and the external bath calibration T_{bath}:

$$\zeta=9/(4\pi L_{max}),\quad J=N\, k_B T_{bath}\ln 2,$$

$$\Delta t_{univ}=h/(m_e c^2\alpha_{PP}),\quad \Delta x_{univ}=c\Delta t_{univ},$$

$$\mu_{M,FPM}=(2/3)\zeta/[(\alpha_{PP}+9)N^4],$$

$$G_{FPM}=\mu_{M,FPM}\zeta c^4\Delta x_{univ}/J.$$

Equivalently, substitution gives $G_{FPM}=(2/3)\zeta^2hc^3/[m_e\alpha_{PP}(\alpha_{PP}+9)N^5k_BT_{bath}\ln 2]$. Under the candidate nine-channel equipartition state law, T_{bath}=306.442 K and the map returns

$$G_{FPM}=6.53990228\times10^{-11}\ m^3\ kg^{-1}\ s^{-2}.$$

Compared with the CODATA value $6.67430(15)\times10^{-11}$ in the same units [13], the relative difference is 2.0137 percent. The nine-channel count is native to the carrier; equal thermal partition is the added physical postulate. No unregistered factor is introduced to eliminate the remaining difference.

5. Time dilation as finite processor lag

Let L_rest be the per-tick action required to maintain a reference process and L the action under gravitational or motion-induced load. FPM identifies their ratio with an effective lag factor

$$\gamma_{FPM}=L/L_{rest}, \quad v_{eff}=c\, L_{rest}/L.$$

The same finite budget therefore links gravitational and motion-induced slowing: additional routing work leaves less causal capacity for internal evolution. Because the action law is bounded, the reference construction has a finite ratio

$$1 \le \gamma_{FPM} \le \gamma_{max}=31.8738629.$$

This ceiling concerns the internal ratio L/L_{rest}. A candidate lag extension introduces a progress accumulator $z_{t+1}=z_t+L_{rest}/L_t$ and identifies a measured clock or decay process with integer crossings of z. A separate motion-to-ledger dictionary would then be required before a particle-lifetime or clock experiment can be registered.

> Status
> The numerical ceiling follows from the selected action bounds. A physical lag prediction requires a declared progress scheduler and an observable map; neither is supplied by the five foundational axioms alone.

6. Cosmology and acoustic structure

6.1 Common boot state

FPM replaces the need for distant regions to negotiate their initial uniformity with a common pre-execution state. A finite substrate can begin from one prepared condition, just as independently addressed storage blocks can share a format because they were initialized together. This is a proposed initial-condition mechanism, not superluminal communication between later regions.

6.2 Acoustic bridge

The numerical audit evaluates a compact algebraic spectrum rather than evolving a complete Boltzmann hierarchy. Its bridge function is

$$S(\ell)=A_{FPM}(\ell/\ell_A)^{n_s-1}\, \exp[-(\ell/\ell_D)^2]\, \{1+A_{osc}\, \sin^2(\ell/\ell_{osc})\}.$$

The fixed quantities include $A_{FPM}=4.0390\times10^{-5}$, $n_s=0.9686263$, tensor ratio r=0.00348596, damping scale $ell_D=1309.5688$, acoustic scale $ell_A=299.82$, and the declared ledger-inertia ratio 16/3. The compact template additionally sets $A_{osc}=0.6$ and $ell_{osc}=80$. Those two oscillation parameters are fixed constitutive template inputs. The scalar local replenishment kernel is a dissipative first-order transfer rule and does not by itself generate acoustic phase oscillations. The displayed spectrum is therefore a phenomenological correspondence, not a derived acoustic-peak prediction; a physical CMB bridge requires additional transfer dynamics and a complete likelihood.

Against the selected Planck comparison [9], the existing fixed-nuisance audit reports Delta chi-squared = +4.16 relative to the Lambda-CDM reference. The positive value means the compact FPM bridge is somewhat worse in that comparison. Its importance is that a low-dimensional route-cost construction reaches the correct qualitative regime and exposes a complete next calculation: implement the transfer functions, polarization, lensing, covariance, foreground nuisance model, and a pre-committed likelihood protocol.

6.3 Tensor suppression

The bridge assigns one specific structural contraction of the nine-channel ledger to the tensor contribution, producing a one-in-nine suppression rule. This is a constitutive channel assignment, not a counting theorem: the 3 by 3 ledger has nine independent entries, while its trace is a contraction of three diagonal entries. A physical tensor derivation must connect that assignment to gauge-invariant perturbations and their propagation.

7. Finite Born distribution

For normalized carrier amplitudes psi_a, FPM uses the Born-compatible weights p_a=|psi_a| squared [10] and realizes them with N finite microcells. Largest-remainder allocation returns integer counts n_a summing exactly to N, with each allocation error below one cell. For nine outcomes and the carrier capacity in Paper 3, the total-variation error is bounded by 3.10×10^{-9}; the reported runtime error is 1.20×10^{-9}.

$$n_a/N \to |\psi_a|^2, \quad D_{TV}<m/(2N).$$

Pure phase rotation leaves the immediate weights unchanged. The current construction establishes finite realization in one selected channel basis. General detector orientation requires an explicit basis-change operation and contextual measurement map; that is the central extension needed before the bridge constitutes a general measurement theory.

8. Joint torsion and Bell/CHSH correlations

8.1 Explicit shared boundary

The FPM pair construction does not assign two independent local carriers prewritten answers. It quantizes a joint distribution across a shared torsion boundary. For detector settings a and b, the ideal correlation is

$$E(a,b)=-\cos[2(a-b)].$$

The four joint probabilities sum to one and leave each wing individually balanced at fifty-fifty. For the standard four settings, the CHSH combination reaches

$$S=|E(a,b)+E(a,b')+E(a',b)-E(a',b')|=2\sqrt{2}.$$

The runtime's finite allocation reproduces the target within its microcell error. An independent local baseline produces the expected triangular correlation and remains at or below S=2.

> Locality classification
> The shared torsion boundary is a non-Bell-local resource. The construction therefore does not evade Bell's theorem or the CHSH formulation [11,12]; it selects a joint architecture outside Bell factorizability and must satisfy no-signalling, relativistic causal consistency, and an independently specified formation and decay law.

8.2 Resource gate

$$q=1/\{1+\exp[10(\max(E_A,E_B)/(0.20E_{max})-0.60)]\}.$$

Its sharpness, midpoint, and scale are constitutive inputs, and it describes joint-resolution activation at low stored energy rather than exhaustion of a paid link. The code's geometric correlation is -cos(a-b); the paper's -cos[2(a-b)] form requires an explicit setting convention, for example $\theta_{code}=2\ \theta_{lab}$.

8.3 Candidate torsion-refresh postulate

A resource-exhaustion mechanism requires an active maintenance operation even when the stored torsion configuration is pure gauge. A candidate postulate is that an experimentally usable shared link executes one refresh transaction per active independent antisymmetric generator per tick. For n_A generators its maintenance cost is $C_{link}=n_A c_0$. With a one-generator link, $c_0=0.05$, and a symmetric available action $A^{(0)}=0.5$ subject to the declared depletion law, the conditional snap threshold is

$$B_*=[A^{(0)}/(n_A c_0)]^{(4/3)}-1=20.5443469.$$

This threshold follows after the refresh postulate and availability map are adopted; it is not a result of the existing zero-cost pure-gauge link. The corresponding Bell transition must specify the laboratory control variable, preparation distribution, and no-signalling constraints.

9. Bare electromagnetic coupling

The torsion-snap bridge uses two distinct floors that must not be conflated. The action floor is $c_0=0.05$. The quantity $e_{floor}=0.0314$ is instead the dimensionless structural percolation floor in the causal-depletion law

$$e_{eff}(B)=\max[(1+B)^{-3/4},e_{floor}].$$

Here $B\geq0$ is the dimensionless local baryonic, or ordinary-matter, load supplied to the viscosity-energy gate; $B=0$ denotes no added matter load. It is a bridge input distinct from cache bias b, bit count B_{erase}, and the action L.

Thus e_{floor} is not c_0/E_{max}, which equals 0.075. With the symmetric-sector exponent beta=9/5, the maximum symmetric occupancy is

$$C_{sym,max}=(1/e_{floor})^{1/\beta},\quad \alpha_{bare}=c_0/C_{sym,max}.$$

Using the shared FPM constants gives inverse alpha_bare = 136.7946397586. The CODATA inverse fine-structure constant is approximately 137.036 [13]; the difference is about 0.1761 percent. FPM interprets its number as a bare coupling before vacuum screening, not as the macroscopic measured value itself. The physical programme is therefore to derive the screening correction from carrier excitations rather than choosing a correction after comparison.

The proposed excitation is a paid, trace-free transverse disturbance released when a shared topological link can no longer be maintained. The decomposition of the 3 by 3 ledger into antisymmetric and symmetric sectors supplies three circulation-like and six strain-like components. Connecting this structure to a massless spin-one gauge field, charge conservation, polarization, and quantum electrodynamic running remains the decisive field-theory derivation.

Status
The bare number is an exact output of the selected upstream constants. Establishing independence from the target requires sensitivity analysis and a derivation whose choices are fixed before comparison with alpha.

10. Candidate extension programs and empirical risk

The bridges are coupled through route cost. This is the framework's central economy and its strongest vulnerability. A failure of the local ledger or viscosity law propagates into gravity, time dilation, cosmology, and the coupling construction; a failure unique to a detector map need not invalidate local conservation or the finite carrier.

- Ledger layer: locality, signed conservation, and finite support are shared by every bridge.
- Constitutive layer: viscosity endpoints and exponents jointly affect gravity, lag, and several reported constants.
- Carrier layer: normalization, phase, consolidation, and finite allocation jointly affect Born, Bell, and Landauer results.
- Nine-channel equipartition extension: an energy-to-route dictionary and equal thermal partition across the nine directed channels determine a Kelvin-valued temperature and propagate directly into G and mass-equivalent scales.
- Torsion extension: a refresh postulate and a control-variable map determine whether a shared link has a resource-dependent survival threshold.
- Lag extension: a progress scheduler and clock dictionary determine whether gamma_FPM maps to a measured rate or lifetime.
- Observable layer: SPARC source functions, cosmological transfer functions, detector bases, and screening maps can fail independently.

10.1 Highest-value next tests

- Predict full two-dimensional galaxy velocity fields and lensing from the tensor ledger, with one frozen parameter set.
- Construct a dynamical cosmological transfer model before treating acoustic structure as a physical prediction.
- Adopt and test a progress scheduler plus a concrete clock or particle observable for gamma_FPM.
- Adopt a torsion-refresh postulate, specify its laboratory resource variable, and test the predicted CHSH transition while preserving no-signalling.
- Derive electromagnetic screening and scale dependence from the same carrier rather than importing the measured correction.

11. Conclusion

FPM's bridge programme is an auditable dependency stack. The foundational ledger supplies local transport, signed accounting, finite allocation, and fixed action bounds. Temperature, physical link maintenance, clock progress, and cosmological transfer dynamics are candidate extension programs with explicit new commitments. Calibrated and correspondence-level numerical comparisons remain distinct from physical predictions.

> Central conclusion
> A physical bridge is valid only at the level warranted by its declared dependencies. The extension programs make their additional assumptions, derived conditional consequences, and failure conditions explicit.

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